
Twenty-one Years of UFO Reports

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My role here today is that of reporter; to report to you on my score or so years of experience with UFO reports (note that I do not say UFO's, for I myself have never had a UFO experience) and with those who make such reports, from this and many other countries. I was asked in 948, as an astronomer then at Ohio State University, and thus geographically near the Wright- Patterson AFB, to review the UFO reports received by the Air Force and to determine how many of them originated from misperceptions of astronomical objects or events. This consulting role continued across the years and gave me the chance to monitor the flow of UFO reports submitted to the Air Force, and to observe the Air Force handling of the problem as first one, then another officer took charge of Project Bluebook.

As reporter of the UFO scene, I am reminded of the old dictum of the reporter: find out Who, What, Where, When, and Why. I will have no difficulty in dealing with the Who, What, Where, and When, for that means simply dealing with facts - particularly with the incontrovertible fact that UFO reports exist, and that the time and location of the reported event is generally known, as well as the identity of the witnesses.

The "Why" I shall leave to other scientists, but I shall challenge their explanations if they are not conversant with the Who, What, Where, and When. I am very weary of pontifications by those who have not done field- or home-work, so to speak.

Indeed, I would like to say a word about scientific methodology as it pertains to this problem. I have discussed this at length with the noted Canadian philosopher of science, Thomas Goudge.

"One of the most interesting facets of the UFO question to me," Goudge writes, "is its bearing on the problem of how science advances. Roughly I would say that a necessary condition of scientific advance is that allowance must be made for (a) genuinely new empirical observations and (b) new explanation schemes, including new basic concepts and new laws." Goudge notes that throughout history any successful explanation scheme, including twentieth-century physics, acts somewhat like an "establishment" and tends to resist genuinely new empirical observations, particularly when they have not been generated within the accepted framework of that scheme - as, for instance, the reluctance to accept meteorites, fossils, the circulation of the blood, and, in our time, ball lighting. History is replete with such examples. When the establishment does accept such new observations it often tends to assimilate them into the going framework

- as, for instance, the attempt to admit the existence of meteorites as stones that had been struck by lightning. "Hence," Goudge concludes, "the present establishment view that UFO phenomena are either not really scientific data at all (or at any rate, not data for physics) or else are nothing but misperceptions of familiar objects, events, etc. To take this approach is surely to reject a necessary condition of scientific advance" [1].

We will never know whether UFO reports represent genuinely new empirical observations if we continue the type of logical fallacy illustrated by the Air Force analysis of a radar-visual UFO report from Kirtland Air Force Base, Albuquerque, New Mexico, in 1957. Two witnesses in the control tower reported at 11:00 p.m. that an object, which looked (through binoculars) like a lighted, up-ended automobile, came within 200 feet of the ground when it disappeared behind a fence in a highly restricted area, easily visible from the control tower, then rose abruptly at very high angular rate and disappeared. It was observed visually for about six minutes, about half of that time through binoculars, and tracked in part by radar. The report of the Air Force officer who investigated this case, which is in the Bluebook file, states:

The two sources are Airways Operations Specialists with a total of 23 years experience. Both were on duty in the control tower at Kirtland Air Force Base when the sighting was made - both appeared to be mature and well poised individuals, apparently of well above average intelligence, and temperamentally well qualified for the demanding requirements of control tower operators. Although completely cooperative and willing to answer any question, both sources appeared to be slightly embarrassed that they could not identify or offer an explanation of the object which they are unshakably convinced they saw. In the opinion of the interviewer, both sources are completely competent and reliable.

Project Bluebook explained this sighting as that of an aircraft; and gave the following specific reasons:

1. The observers are considered competent and reliable sources and in the opinion of the interviewer actually saw an object they could not identify.
2. The object was tracked on a radar scope by a competent operator.
3. The object does not meet identification criteria for any other phenomenon.

So, the witnesses were solid, the radar operator competent, and the object unidentifiable as any other phenomenon; therefore the object had to be an aircraft. Clearly, if such reasoning is applied to all UFO reports we can hardly expect to find out whether any genuinely new empirical observations exist to be explained. Schroedinger, the father of quantum mechanics, wrote: "The first requirement of a scientist is that he be curious; he must be capable of being astonished, and eager to find out." Perhaps he should have added, "and be ready to examine data even when presented in a bewildering and confusing form."

There is much in the UFO problem to be astonished about - and much to be confused about, too. Such confusion is understandable. Over the past twenty years I have had so many experiences with crackpots, visionaries, and religious fanatics that I hardly need be reminded of people who espouse the idea of UFO's as visitors from outer space for their own peculiar purposes. You will note that I say "espouse the idea," not "make UFO reports." Very rarely do members of the lunatic fringe make UFO reports. There are many reasons for this; primarily it is simply that they are incapable of composing an articulate, factual, and objective report.

In addition to being fully aware of the cultists, and how they muddy the waters even though they don't generate UFO sightings, I am also well aware of the widespread ignorance, on the part of many, of astronomical objects, high-altitude balloons, special air missions, mirages, and special meteorological effects, and of people's willingness to ascribe their views of such things to the presence of something mysterious. These people, in contrast to the crackpots, are far more of a problem because they do generate UFO reports

which represent a high noise level - so high, in fact, that many who have not looked carefully into the matter feel that all UFO reports stem from such misperceptions. In actual fact it is relatively simple for an experienced investigator to sort out and quickly eliminate virtually all of the misperception cases.

It is a pity that people so often are not well-informed, objective, and accurate reporters; since 1948 I have become only too familiar with UFO reports spawned by Venus, twinkling stars, aircraft, and the like. Some eighteen years before the Condon committee was formed I was already aware that the great majority of UFO reports are nothing more than misperceptions by the uninformed. Of course, these misperceptions must be deleted before any serious study of the UFO question can begin. From this point on, I am speaking only of UFO reports which remain unexplained by trained investigators; only then are we truly dealing with something that is unidentified by people capable of making an identification. In short, an original UFO report must pass through a "narrow band-pass filter" before it qualifies as worthy material for scientific study, the objective of which is to determine whether any genuinely new empirical observations exist. Only those reports which survive the running of this gauntlet can qualify. An objection to this approach immediately arises: Aren't we just rejecting everything but the tail-end of the distribution curve of human reactions to visual stimuli? I firmly agreed with this view (during my first years of association with the UFO problem, but now I question it. We can take the position that we are dealing with the vagaries of human perception only if we are dealing with a homogeneous set of observations. For instance, the distribution curve of fruit size in an apple orchard would have a significant tail at the large-fruit end if measurements of watermelons on the ground were included without noting the structural differences between apples and watermelons.

Let me define the UFO phenomenon, the *existence* of which we wish to determine or deny, as that phenomenon described by reports of visual or instrumental observations of lights or objects in the sky (or near, or on the ground) whose presence, trajectories, and general character are not explainable in verifiable physical terms, even after intensive study. The Condon Report furnishes us with many examples [2].

For years I could not accept the idea that a genuine UFO phenomenon might exist, preferring to hold that it was all a craze based on hoaxes and misperceptions. As my review of UFO reports continued, and as the reports grew in number to be of statistical significance, I became concerned that the whole subject didn't evaporate as one would expect a craze or fad to do. Also, the phenomenon of UFO reports not only persisted in this country but in many areas over the world; if there were some worldwide compulsion to report strange things, why are only these particular types of strange reports preferred from the infinite universe of all possible strange reports? The degree of "strangeness" is certainly one dimension of a filtered UFO report. The higher the strangeness index the more the information aspects of the report defy explanation in ordinary physical terms. An other significant dimension is the probability that the report refers to a real event; in short, did the strange thing really happen? And what is the probability that the witnesses described an actual event? This credibility index represents a different evaluation, not of the report in this instance, but of the witnesses, and it involves different criteria. These two dimensions can be used as coordinates to plot a point for each UFO report on a useful diagram. The criteria I have used in estimating these coordinates are: For strangeness: How many individual items, or information bits, does the report contain which demand explanation, and how difficult is it to explain them, on the assumption that the event occurred? For credibility: If there are several witnesses, what is their collective objectivity? How well do they respond to tests of their ability to gauge angular sizes and angular rates of speed? How good is their eyesight? What are their medical histories? What technical training have they had? What is their general reputation in the community? What is their reputation for publicity-seeking, for veracity? What is their occupation and how much responsibility does it involve? No more than quarter-scale credibility is to be assigned to one-witness cases. If one now plots the strangeness (S) of a report against the credibility (P) of the witnesses - that is, the probability that the event happened more or less as stated - one obtains a diagram which may be called the strangeness-probability diagram. An example of such a diagram constructed for some cases I have personally investigated is shown in Figure 4-1. Plotted points represent only those UFO

reports, of course, that have passed through the misperception and hoax filter. Clearly, the most provocative and potentially important UFO reports are those in the upper right-hand region of such a diagram, representing reports that contain many information elements and have a high probability rating.

In these high-P reports, the witnesses were of such a caliber and the circumstances surrounding the reported event were such that we cannot discount the reported event. Examples of such information bits are craft description, motions that seemingly defy inertial laws, effects on animals, interference with automobile ignition systems, and visible marks on land. The Condon Report includes several such cases.

My long experience in personal contacts with witnesses who generate high-S high-P reports shows that all were trying to describe an event for which they had an entirely inadequate vocabulary - much as an aborigine lacks the vocabulary to describe a supersonic jet or a nuclear submarine. Whatever else can be said of the UFO phenomenon, it represents for the witness an undoubted event, and an event for which he was totally unprepared. The majority of such witnesses, contrary to popular belief, were originally highly skeptical about UFO's. Suddenly they had an experience which affected them profoundly, sometimes traumatically. Faced with the experience of the UFO event, witnesses are generally perplexed and uncertain as to what to do about it. Invariably they attempt to explain it in ordinary terms and fail. Curiosity overwhelms them yet they know that they will be targets for ridicule if they report (they confess that they had often in the past ridiculed others). Generally they first confide only in their own family, and often prefer to remain silent. Only those who finally report their observations furnish us with data for study.

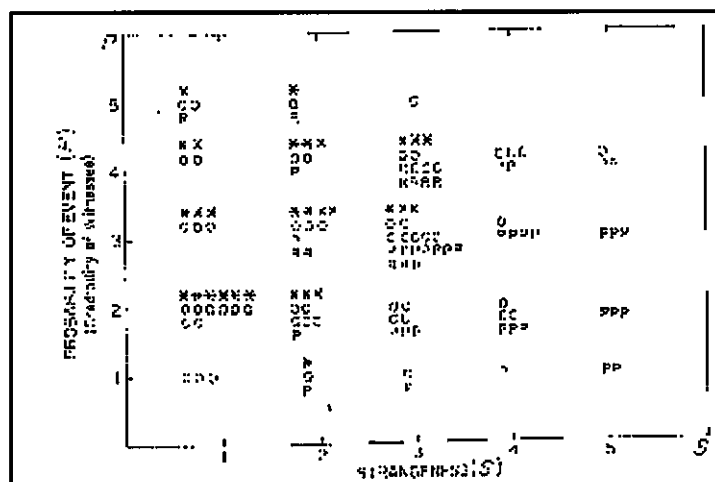


Figure 4-1. strangeness/probability diagram of UFO sightings. To be considered important, such a sighting must have both a very high probability of having actually occurred and a very high strangeness. The upper right-hand corner of the diagram is not heavily populated. * = nocturnal lights; O = daylight disks; R = radar cases; C = close encounters with no interaction with the environment; P = close encounters with physical effects (landing marks, burnt rings, engine stoppages, etc.).

Any serious investigator is aware that many unreported experiences must exist. Not only has a Gallup poll so indicated, but I frequently try the experiment of asking for a show of hands of those who either have had a puzzling UFO experience themselves or have heard of one from close friends. I generally find that more than 10 per cent of the audience will raise a hand. But when I ask for hands of all those who reported the event in some official manner, I find virtually no hands raised. Judging from this and other personal observations, I would estimate that for every officially reported UFO sighting there may exist dozens that have gone unreported. As scientists we should be astonished that high-S high-P reports have been made in the past five or ten years. What does a serious person with a valued reputation stand to gain by making such a report? Why do people go to the trouble of filling out questionnaires, of subjecting themselves to sometimes hostile inquiry, and of being the target of unpleasant attention?

The reason appears to be twofold. Witnesses have told me that they had not intended to say anything about their experience but they felt that it might be of importance to the government, or to science, and felt it their duty to report. The second reason is curiosity. They want to know whether anyone else experienced the same event, and whether the event has a rational explanation. They are visibly reassured when I tell them, if it be the case, that their sighting fits a pattern and resembles other reported sightings from various parts of the world.

What are the patterns of UFO reports? How can we classify UFO reports (after screening) as an aid to their study? Clearly, if each UFO report represents a unique happening, the UFO is not amenable to scientific study. Such a classification, however, must be free of any preconceived ideas as to the nature and cause of UFO's. Thus the classification must be descriptive; it should be similar to the classification of stellar spectra in the days before we had a theory of stellar spectra, or somewhat like the classification of galaxies today.

I have adopted a very simple classification system based solely on the manner of observation. Such a system tells us nothing, of course, about the nature of the UFO, but it can suggest a means of gathering further data. There seem to be four basic ways in which the UFO presents itself, so to speak, for human observation: (1) as "nocturnal lights," the objects to which the lights are presumably attached being generally barely, if at all, discernible, (2) as "daylight disks," when the UFO generally, though not necessarily, appears as a disk or long oval; (3) as "close encounters" during day or night: these are sightings made at ranges less than 1,000 feet and often accompanied by physical effects on the land, on plants and animals, and occasionally on humans; and (4) radar UFO's, a special subset of which is the radar-visual observation, in which the radar and visual observations are mutually supporting. These observational classifications are not meant to be mutually exclusive. Clearly a nocturnal light might be a daylight disk in the daytime, and both might become close-encounter or radar cases.

Let us examine each category. A nocturnal-light report offers the least potential for scientific study, as it has the fewest information elements and thus a low strangeness index. The nocturnal-light UFO can be defined as a light or combination of lights whose kinematic behavior passes through the "UFO report filter"; that is, it cannot be logically ascribed to balloons, aircraft, meteors, planets, satellites, satellite reentries, or missiles. The experienced investigator generally has no difficulty with the screening process here. Years of checking enable him to filter these out almost at first glance. Of course, should a UFO choose to masquerade as a hot-air balloon or a photographic night-air exercise, there is no easy way of differentiation, at least so long as we are limited to observing from the ground. If we had immediate reaction capabilities, and could dispatch an interceptor aircraft, then we could clear the matter up quickly, or perhaps we would experience what has often been reported in the past twenty years: as the intercepting plane approaches the light in question, the light either suddenly goes out or seems to take off and soon outdistances the investigator. In that event the report earns its place as a high-S high-P member of the nocturnal-light category.

An example of this category is a case I investigated personally, involving five witnesses, the senior witness being the long-time associate director of a prominent laboratory at MIT. The nocturnal light was first sighted by his son, who had been out airing the dogs. He came bounding into the house crying, "There's a flying saucer outside!" The senior observer picked up a pair of binoculars on his way out. He told me that he didn't expect to see anything unusual but was going out to see what the commotion was all about. For the following ten minutes he was engrossed in what he saw - the nature of the light, its motions, its hovering, and its take-off. He described the light as having a high color-temperature although essentially a point source, subtending less than a minute of arc in the binoculars. The five observers were fortunately able to compare it to an airliner and a helicopter, both of which passed by during the observation interval, and neither the motions nor lights of these craft bore any resemblance to those of the UFO, subclass NL. The trajectory of the object was plotted against the framework of the branches of a denuded tree. This observer

was a good one, and his report included the condition of his eyes and those of the members of his family. The adult observers were both far-sighted and the senior observer wore glasses only for reading.

Incidentally, all my attempts as scientific consultant to the Air Force, to mount a serious investigation of this case came to naught. The Bluebook evaluation is "unidentified," but somehow this word is not a challenge to inquiry. It has been classified as "unidentified," and therefore the case is "solved": it has been identified as "unidentified"! So certain is the Air Force, at least publicly, that all UFO reports must represent normal things that they see no point to serious investigation. During most of the time I acted as their consultant I repeatedly urged immediate reaction capability and proper scientific investigation, but to no avail.

The second category, the "daylight disk," covers reported daylight sightings of objects seen at moderate distances. The prototype report runs something like this: I was driving along and there crossed over in front of me a shiny metallic disk. It seemed about 500-1,000 feet above the road. It came down fairly close to the ground, stopped and hovered with a wobbling motion and then took off with incredible speed, straight up, and was gone in a few seconds. There was no noise. This category understandably has more photographs to support it than all the others put together. An example is the McMinnville, Oregon, case which the Condon Report lists as unsolved (Case 46).

A photographic daylight disk case was reported by three prospectors in bush country near Calgary, Alberta. I personally investigated the terrain, the people, the negatives, and the camera. Fred Beckman of the University of Chicago and I have satisfied ourselves that the images on these color negatives are real images. The terrain, the interrogations of the witnesses, and the sworn affidavit of the principal witness all lead me to ascribe a high SP rating to this case.

The published literature on UFO's has many photographs. Some are clearly hoaxes, but many have never been investigated sufficiently to rule out very sophisticated hoaxes. A hoax is all one has to rule out, however. For if the daytime photo shows any detail at all, aircraft, balloons, and so forth may be immediately eliminated. The picture itself is sufficient to establish the Strangeness index; it is the credibility index that is difficult to assess. Proper interrogation, tracing of the processing history of the negative, microscopic and microphotometric examination of the negative, plus proper psychological testing of the witnesses to the taking of the photograph should serve to rule out all but the most highly sophisticated, expensive, and laboriously contrived hoaxes. In any one case it is clearly impossible ever to state unequivocally that a photo of a daylight disk is genuine, but I would submit that twenty-five such separate photographic cases, each subjected to exhaustive tests, would allow us to say that the probability of a hoax in all twenty-five cases is vanishingly small.

Even this does not prove the existence of truly strange flying objects, but it should be sufficient to attract the proper attention of the scientific world. That, of course, has long been my position: that some UFO reports are worthy of serious scientific attention. Inherent in the sheaves of UFO reports may well be many doctoral dissertations for physicists, sociologists, and psychologists alike.

The third category of UFO reports, the "close encounter," offers by far the greatest potential for scientific study. Since a close encounter obviously offers a greater chance for observation, we can expect, and we get, many more information elements, and hence a higher strangeness index. Here the theory of simple misperception fails utterly in explaining reports of craft landing 100 feet away, of visible marks left on the ground, of animals and people visibly affected, and of automobiles temporarily stopped on the road. Either we must say that the witnesses were mentally unbalanced or that something most interesting actually happened. However, I am not taking sides; I am merely reporting to you what is reported all around the world, and by seemingly competent witnesses.

I divide the close encounter cases into three subdivisions: the close encounter, with little detail; the close encounter with physical effects; and the close encounter in which "humanoids" or occupants are reported. This latter subgroup, of course, has the highest strangeness index and frightens away all but the most hardy investigators. I would be neither a good reporter nor a good scientist were I deliberately to reject data. There are now on record some 1,500 reports of close encounters, about half of which involve reported craft occupants. Reports of occupants have been with us for years but there are only a few in the Air Force files; generally Project Bluebook personnel summarily, and without investigation, consigned such reports to the "psychological" or crackpot category.

A prototype of the simple close encounter goes like this: Witnesses are driving along a lonely road when the driver spies a strange glare in his rear-view mirror. He becomes frightened and increases his speed, trying to outdistance the UFO, but he cannot. He stops the car and tries to take cover. Shortly the light rises and vanishes quickly in the distance. It is easy to say that such witnesses are mentally unbalanced, unless one must say it to their faces, especially knowing that they are respected members of their communities and hold responsible positions.

The close encounter with physical effects is the category that interests me most, since the reported effects on animal, vegetable, and mineral are potentially measurable. For instance, there are more than a hundred reports on record of UFO's that reportedly caused car ignition failures. In a typical case a bright light suddenly appears and seems to seek out the witnesses' car. As it stops to hover over the car, the car lights dim, or fail, and the engine dies. Often the occupants of the car report feeling hot and prickly. After a few minutes the apparition leaves, and the car returns to normal operation, but the witnesses often do not; their equanimity is temporarily destroyed.

Witnesses of such encounters do not readily submit themselves to interrogation. Often they tell no one for days, or they tell only very close associates. Eventually a serious UFO investigator comes to hear about it, and then the story unfolds. If they unwisely tell their story indiscriminately, their lives are made miserable by ridicule and taunts of unsympathetic friends.

Let us consider the probabilities in car-failure cases. On the road we occasionally pass a disabled car, its hood up, waiting for the repairman or the tow truck. We should regard it as odd, and of low probability, if the car were to heal itself after a few minutes and proceed as though nothing had happened. If we add that this event was accompanied by a very bright unexplained light hovering over the car, then I submit that the probabilities are extremely small. And when we deal with dozens of such cases, we are driven to the conclusion that something most extraordinary must have happened. If we have in these cases what Goudge calls "genuinely new empirical observations [requiring] new explanation schemes," then we can anticipate an exceptional scientific breakthrough, although it may not be just around the corner.

In this twentieth century we may be as far from a solution of the UFO problem as nineteenth century physicists were from an interpretation of the aurora borealis. Even so, it is incumbent upon us as scientists to document and study the phenomenon to the best of our ability. But the present lack of continued scientific study still leaves it unclear whether genuinely new empirical observations exist. Even the Condon Report left unexplained about one-quarter of the cases examined.

The fourth observational category contains those UFO reports involving radar. There are many reports in this category from responsible persons, such as pilots and control tower operators. I have paid little attention to the radar cases, since I am not a radar expert. The expert on Project Bluebook invariably ascribed all radar cases to malfunction or to anomalous propagation. [*] The Condon Report, however, contains the following remark about one such case: "This must remain as one of the most puzzling radar cases on record and no conclusion is possible at this time. It seems inconceivable that an anomalous propagation echo would behave

in the manner described, even if anomalous propagation had been likely at the time" [3]. Radar-visual cases offer more scope for study. The Lakenheath case (see Chapter 5, Appendix, Case 2), studied by the Condon committee, was left as unexplained with the remark, "In summary, this is the most puzzling and unusual case in the radar-visual files. The apparently rational, intelligent behavior of the UFO suggests a mechanical device of unknown origin as the most probable explanation of the sighting. However, in view of the inevitable fallibility of witnesses, more conventional explanations of this report cannot be entirely ruled out" (p. 164). In actuality, a careful reading of the body of the Condon Report reveals as good a case for the scientific study of UFO's as could have been assembled by a group not initially conversant with the subject and with limited time and funds.

[*] The latter arises when meteorological conditions are such as to interfere with the normal straight-line propagation of radar waves, leading to erroneous interpretation of the radar results.

Some may be surprised that so considerable a body of UFO evidence exists. This is the crux of the problem: neither the active scientists nor the public have access to this information. Unfortunately, those who wish to learn about UFO's must get information from the "back fences" of literature - the pulp magazines, the sensational mystery or sex magazines. Until recently, there has not been in this country one scientific journal in which I could publish a well-documented UFO case, yet a recent bibliography of UFO literature ran to 400 pages. The UFO has become a problem for the librarian sooner than it has for the scientist.

Consider the plight of serious UFO witnesses. I know that such exist because I have interviewed several hundred. Where can they go to report? Only the most naive would report to the Air Force even if Project Bluebook had not been discontinued. To report to the local police is scarcely better. Many witnesses have told me of the ridicule they met when they took that path. Besides, I have seen many police blotters. UFO reports are entered as "complaints."

The witness, if he wishes to report, must seek out the relatively few persons or organizations which will lend a sympathetic ear. My own mail brings me very good UFO reports, generally with a request for anonymity, but I have neither the time nor the funds to make proper investigations. As I look back over my past twenty-one years' association with the UFO problem, I note that the intellectual climate today is enormously better for taking a good look at it than it was even a few years ago. This symposium is itself an example: it would have been impossible to have held it a year or two ago. And had I, earlier, attempted to call for a major investigation, I would have lost credibility and undoubtedly all possible future effectiveness.

In summary, then, my twenty-one years of monitoring of UFO reports has shown that reports of UFO observations remain after we delete the pronouncements of crackpots, visionaries, religious fanatics, and so forth. A large number of UFO reports are readily identifiable by trained investigators as misperception of known objects and events. A small residue of UFO reports are not so identifiable. These come from such widely separated places as northern Canada, Australia, South America, and Antarctica. They are made by competent, responsible, psychologically normal people - in short, credible witnesses. These reports contain descriptive terms which collectively do not specify any known psychological event, object, or process, and which do not specify any known psychological event or process. And, furthermore, they resist translation into terms that do apply to known physical and/or psychological events, objects, and processes. That is, as Goudge points out, translation would alter the meaning of the original report and hence effectively violate the methodological criteria governing the advance of science: namely (a) that it must be possible for new observational data to occur; that is, the existing conceptual framework of science, or the attitudes of scientists, must not rule out such new data a priori; and (b) the existing conceptual framework must allow new concepts, principles, and laws to be formulated to interpret and explain the new observational data.

Although I know of no hypothesis that adequately covers the mountainous evidence, this should not and

must not deter us from following the advice of Schroedinger: to be curious, capable of being astonished, and eager to find out.

Notes

1. Thomas Goudge, personal letter to author.
2. E. U. Condon, Scientific Study of Unidentified Flying Objects (New York: Bantam Books, 1969).
3. Ibid., p. 171.
4. Ibid., p. 164.
5. Lynn Catoe, UFO's and Related Subjects: An Annotated Bibliography. Library of Congress, AFOSR 68-1656 (Washington, D.C.. U.S. Government Printing Office, 1969).

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